

COMMODITIES MARKETS AND MODELS : REPORT

1. Introduction

This project aims to develop quantitative trading strategies applied to commodity markets, with a focus on the sugar sector. The objective is to build a structured dataset and use statistical methods to identify trading opportunities.

The dataset includes a commodity anchor (sugar futures, $SB=F$), several sector ETFs (MOO, DBA, CANE), and a set of equities involved in the sugar value chain, such as producers, processors, and food companies. This structure allows us to capture both the overall market dynamics and the specific impact of sugar prices on related assets.

The sugar market belongs to the agricultural commodities sector, where prices are mainly driven by supply and demand. Production depends on factors like weather conditions and major producing countries such as Brazil and India, while demand is linked to global food consumption and industrial uses. This makes sugar prices relatively volatile. Because of these economic links, some equities may share a long-term relationship with sugar prices. This can lead to cointegration, meaning that even if prices move differently in the short term, they tend to return to a common equilibrium over time.

In this project, we use these relationships to build trading strategies, especially a pairs trading approach that exploits mean-reversion between sugar futures and the most related equity.

2. Data Collection & Universe Construction : G18 Sugar

2.1 Asset Universe

To build our asset universe, we followed the three-tier structure described in the project guidelines.

As the primary anchor, we picked the **Sugar #11 Futures contract ($SB=F$)**, which is the main international benchmark for raw sugar. This is the series we use as the core commodity reference throughout the strategy.

For the sector ETFs, we kept three. The **Teucrium Sugar Fund (CANE)** is the most direct one since it directly tracks sugar futures. The **Invesco DBA** gives broader exposure to agriculture in general, and **VanEck's MOO** captures more of the agribusiness equity side. We had also included SGG (iPath Bloomberg Sugar ETN) initially, but it got dropped during cleaning because it had too many missing observations.

For the individual stocks, we selected 14 listed companies with direct or indirect exposure to sugar: agricultural commodity traders like **Bunge (BG)** and **ADM**, and large food & beverage groups that are

heavy sugar consumers like Hershey, Mondelez, PepsiCo, Coca-Cola, or Kraft Heinz. The idea is that their margins and stock prices are sensitive to sugar price movements, which makes cointegration tests against SB=F meaningful.

One note on **Cosan SA (CSAN)**: it's normally the reference Brazilian sugar producer in the G18 list, but we had to drop it because Yahoo Finance had too many gaps over our study period, likely due to restructuring events and trading halts on the Brazilian exchange (B3). With 14 other assets remaining, this doesn't affect the statistical robustness of the analysis.

Several mandatory tickers from the G18 list were excluded for data quality or availability reasons: TATE.L and ABF.L (London-listed, significant calendar mismatches with US futures), WILMAR (Singapore-listed, insufficient liquidity on Yahoo Finance OTC), and SRE (Sempra Energy, whose business model has no material exposure to sugar prices). These were replaced by economically coherent substitutes from the sugar value chain: KHC, KO, GIS, MKC, CPB, PEP, KDP.

2.2 Data Source & Period

All data was downloaded using the Python library `yfinance`, with `auto_adjust=True` to get prices already adjusted for dividends and splits. We covered the period from **January 1, 2019 to December 31, 2024**, which gives us 6 full years of daily data — above the 5-year minimum required.

2.3 Data Cleaning

The cleaning process was pretty straightforward and done in three steps. First, we removed any ticker with more than 20% missing values — that's where CSAN and SGG were dropped. Then we applied a forward-fill to handle small gaps coming from public holidays and calendar mismatches between futures and US equities. Finally, we dropped the few remaining rows with NaNs, which usually appear at the very beginning of the sample before all series are active.

The final cleaned panel contains **18 assets across 1,510 trading days**, from January 2019 to December 2024. We saved the full OHLCV dataset as `sugar_dataset_G18.csv` and the adjusted close panel as `sugar_prices_panel.csv`, both included in the submission.

3. Trading Game #1

3.1 Multivariate Statistical Arbitrage

In this first trading game, we implement a multivariate statistical arbitrage strategy applied to the sugar market. The objective is to construct an optimal market-neutral basket of equities that captures the common dynamics of assets linked to sugar.

We start by selecting the equities that are the most cointegrated with the sugar futures (SB=F), using the Engle-Granger test. We retain the five assets with the lowest p-values, as they exhibit the strongest long-term relationship with the commodity.

To estimate this relationship, we run an Ordinary Least Squares (OLS) regression where sugar futures are regressed on the selected equities. The estimated coefficients represent hedge ratios and define the equilibrium between the assets.

We then build an optimization model using Gurobi. The objective is to minimize the variance of the basket returns, allowing us to construct a risk-efficient portfolio. Two constraints are imposed:

- A market-neutral constraint, ensuring that the sum of portfolio weights is equal to one.
- A constraint linking the portfolio weights to the OLS betas, in order to preserve the cointegration structure.

3.2 Spread Construction and Trading Strategy

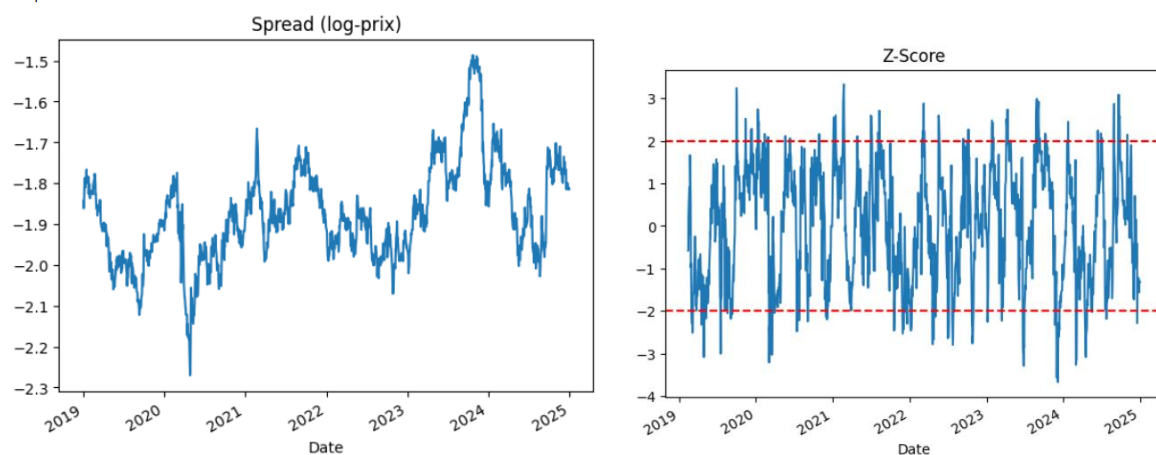
Once the optimal weights are obtained, we construct the basket as the weighted sum of the selected equities. The spread is then defined as the difference between sugar futures (SB=F) and the basket.

To identify trading opportunities, we compute a Z-score of the spread using a 30-day rolling window. This normalization allows us to see deviations from the equilibrium.

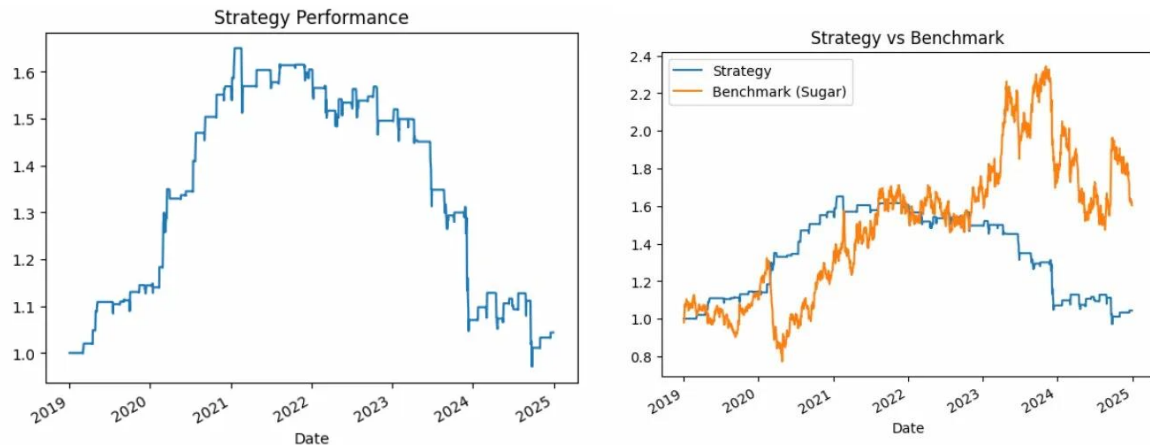
The trading strategy follows a mean-reversion logic:

- A long position is taken when the Z-score falls below -2, indicating that the spread is undervalued.
- A short position is taken when the Z-score exceeds +2, indicating that the spread is overvalued.
- Positions are closed when the spread starts reverting toward its mean.

ADF p-value : 0.0156 → stable



3.3 Results and Performance Analysis



The multivariate strategy generates a Sharpe ratio of 0.12, which indicates a very low level of risk-adjusted performance. While the strategy does not consistently lose money, its returns are not sufficient to compensate for the level of risk taken.

The maximum drawdown reaches -41.19%, which is relatively high for a market-neutral strategy. This suggests that the model is exposed to significant adverse movements and that the spread does not always revert quickly to its equilibrium.

The Calmar ratio is 0.11, confirming the weak performance of the strategy. The combination of low returns and large drawdowns highlights an unfavorable risk-return trade-off.

One key observation is that the spread does not exhibit strong or consistent mean-reverting behavior. This can be explained by the fact that combining multiple equities dilutes the cointegration relationship observed at the individual level. As a result, the signals generated by the Z-score are less reliable and often leads the program to take trades that are not that profitable

Additionally, the optimization constraints may limit the flexibility of the portfolio. By imposing both market neutrality and consistency with OLS betas, the model may produce suboptimal weights that do not fully capture short-term deviations in the spread.

Compared to the pairs trading strategy developed in Game #2, the performance is significantly weaker. This confirms that a 1vs1 approach is more effective in capturing strong cointegration relationships, while the multivariate strategy tends to smooth out both risk and return.

Overall, the results show that although the multivariate statistical arbitrage framework is theoretically a good strategy, it seems that its practical implementation in this context is not efficient. The strategy lacks strong mean-reversion signals and fails to deliver attractive performance.

3.4 Conclusion

This first trading game explored the implementation of a multivariate statistical arbitrage strategy on the sugar market. The approach combined cointegration analysis, OLS-based hedge ratio estimation, and portfolio optimization to construct a market-neutral basket of equities.

While the theoretical framework is reliable, the empirical results show that the strategy struggles to generate strong and consistent returns. The low Sharpe ratio (0.12) and the high maximum drawdown (-41.19%) indicate that the model does not effectively capture reliable mean-reversion opportunities in this multivariate setting.

One of the main limitations comes from the diversification effect. By combining several equities into a single basket, the strength of individual cointegration relationships is diluted, which reduces the quality of the trading signals. In addition, the optimization constraints, although necessary to ensure market neutrality and consistency with the OLS regression, all of this leads to suboptimal weight allocation and so poor performances

Despite these limitations, this approach remains useful from a risk management perspective, as it provides a structured way to build diversified and market-neutral portfolios. It also highlights the importance of carefully selecting assets and designing constraints when implementing multivariate strategies.

Overall, this trading game provides valuable insights into the challenges of multivariate statistical arbitrage. It shows that while diversification can reduce risk, it may also weaken the strength of trading signals, leading to lower performance compared to simpler strategies such as pairs trading.

4. Trading Game #2

4.1 Cointegration Analysis

In this part, we try to identify which stock has the strongest long-term relationship with sugar futures ($SB=F$), since our trading strategy is based on mean reversion.

First, we check whether the price series are stationary using the Augmented Dickey-Fuller (ADF) test. As expected, most of the price series are non-stationary in levels but become stationary after differencing. This means they are integrated of order one ($I(1)$), which is a necessary condition before testing for cointegration.

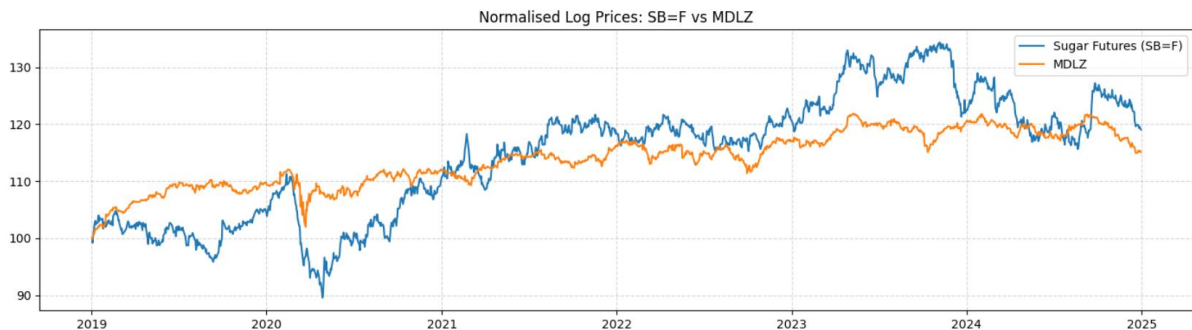
Then, we apply the Engle-Granger cointegration test between sugar futures and each stock in our dataset. The goal is to see whether a linear combination of the two series is stationary, which would indicate a stable long-term equilibrium relationship.

The results show that several stocks are cointegrated with sugar futures at the 5% significance level, including MDLZ, KHC, PEP, and BG. Among them, Mondelez (MDLZ) has the lowest p-value (0.0037), which suggests the strongest cointegration relationship.

To confirm this result, we also run a Johansen test on the top candidates. The test indicates the presence of at least one cointegrating vector, which reinforces the idea that these assets share a common long-term trend.

Based on these results, we select the pair $SB=F$ and MDLZ for our trading strategy. This pair is expected to exhibit mean-reverting behavior, which we will exploit using a Z-score-based trading strategy in the next step.

4.2 Visual Analysis of the Pair



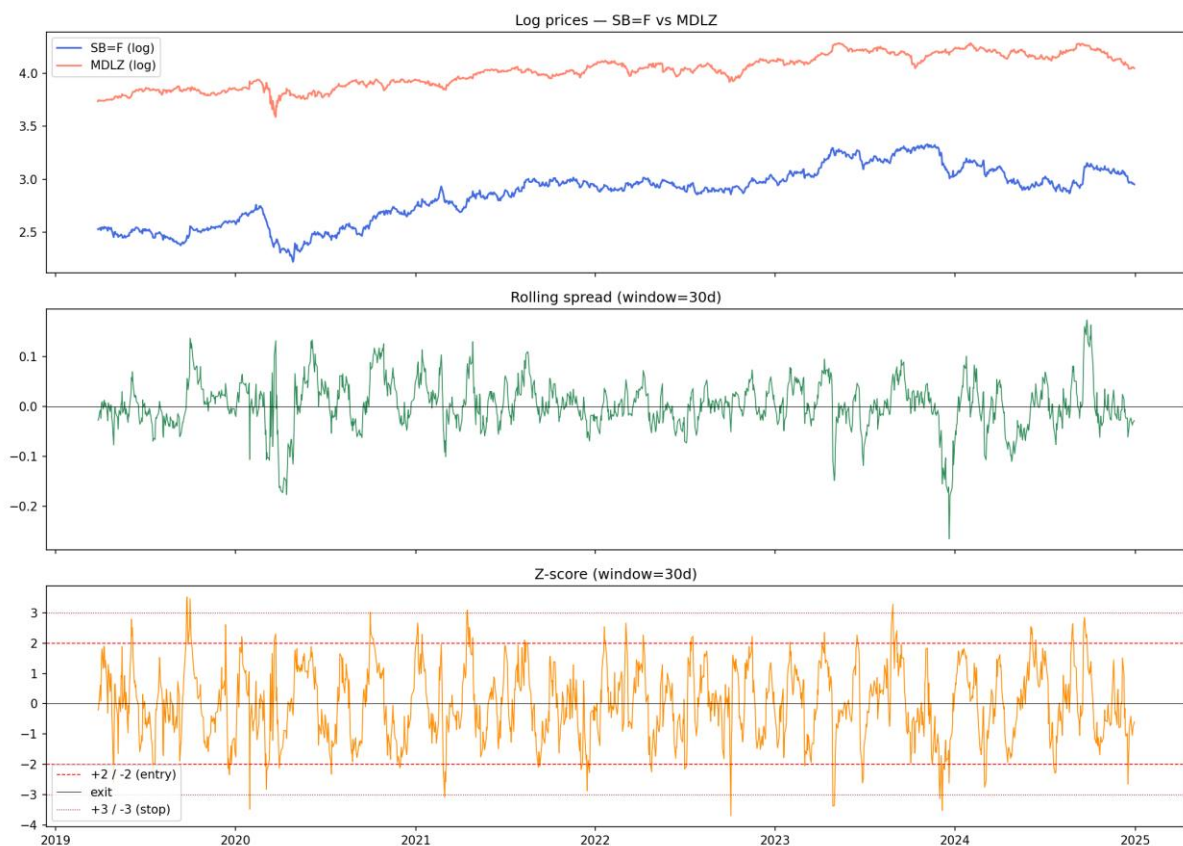
The figure shows the normalized log prices of sugar futures (SB=F) and Mondelez (MDLZ). Both series follow a similar long-term trend, even if short-term movements differ. We can observe several periods of divergence, especially during market stress. These deviations are important, as they suggest potential mean-reversion opportunities.

Overall, this visual evidence is consistent with the cointegration results and supports the choice of this pair for the trading strategy.

4.3 Z-score Construction and Lookback Selection

To build the trading signal, we first compute the spread between sugar futures and the selected equity using a rolling OLS regression. This allows the hedge ratio to vary over time instead of remaining fixed, which makes the strategy more adaptive to changing market conditions.

We then test several lookback windows (30, 60, 90, 120, and 180 days). For each window, we measure the stationarity of the resulting spread using the ADF test. The 30-day window gives the lowest p-value and is therefore selected as the optimal lookback.



The final Z-score is computed from the rolling mean and standard deviation of the spread. Its mean is very close to zero, and the series shows several significant deviations above and below the entry thresholds, which suggests that the pair provides exploitable mean-reversion signals.

4.4 Trading Strategy & Backtest Results

The strategy trades the spread between SB=F and MDLZ based on Z-score signals. We go long the spread when it drops below -2 and short when it exceeds +2, exiting when it crosses back through 0.

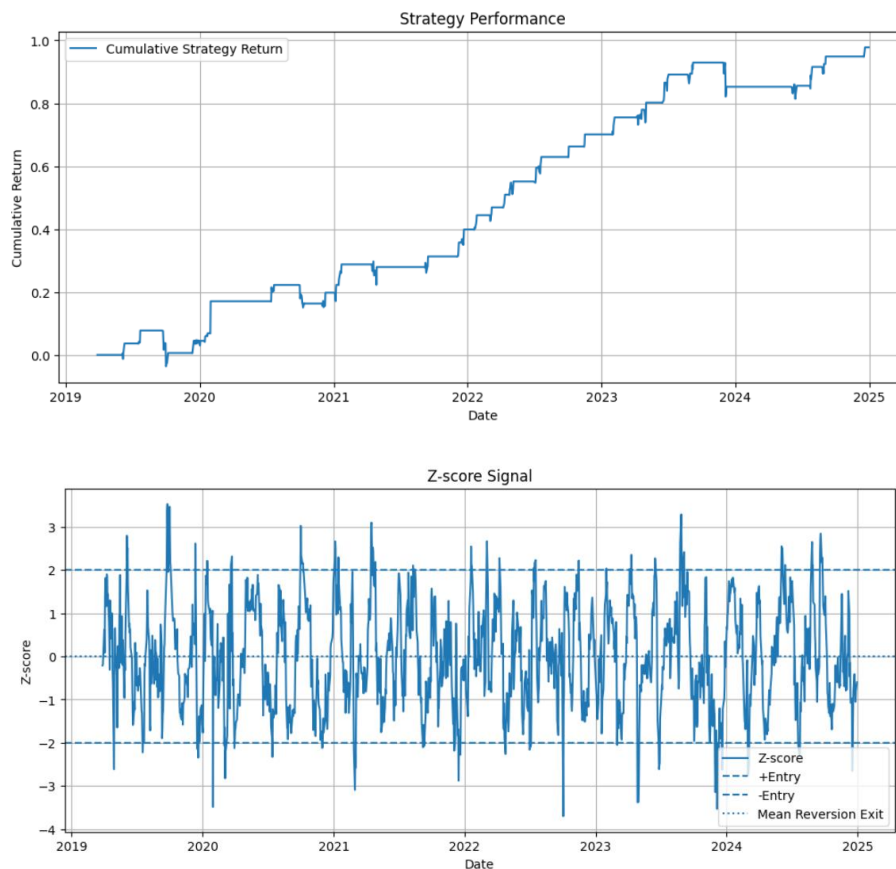
To improve risk management, we implement an adaptive trailing stop based on the Z-score dynamics. Once a position is opened, the strategy tracks the most favorable Z-score reached during the trade. For a long position, the stop is defined as $\text{best_z} + 1.5$, while for a short position it is defined as $\text{best_z} - 1.5$. The position is closed if the Z-score crosses this dynamic threshold, allowing the strategy to protect gains and limit adverse movements.

We added two additional risk controls. First, a volatility filter that blocks new entries when the 20-day rolling volatility of SB=F is above its 75th percentile — this kept us out of the market for about 370 days, mostly during the COVID shock in 2020 and the sugar rally in 2022–2023. Second, transaction costs of 10bps per trade, as required by the project guidelines.

Over the 2019–2024 period, the strategy generated 37 trades, which is roughly 6 per year — a reasonable frequency for a mean-reversion approach on a commodity spread. The cumulative return is 0.98 over 6

years, with a Sharpe ratio of 1.35, a max drawdown of -11.6%, and a Calmar ratio of 8.46. The win rate is 83.8%, which is consistent with mean-reversion strategies that rely on frequent small gains and controlled losses.

Overall, the results are strong. The volatility filter clearly improved the risk profile, reducing drawdowns and avoiding trading during stressed market conditions.



4.5 Conclusion

This project implemented an optimized 1-vs-1 pairs trading strategy on the sugar commodity market, following the Game #2 framework. The full pipeline went from data collection to backtesting, covering all the steps required by the guidelines.

The most interesting finding was the cointegration result. Out of 14 candidate equities, Mondelez International (MDLZ) emerged as the best pair with Sugar #11 Futures (SB=F), with an Engle-Granger p-value of 0.004. This result is economically meaningful, as Mondelez is one of the largest sugar buyers in the world through its chocolate and biscuit products. Therefore, the relationship is not purely statistical but also supported by a strong fundamental link.

The dynamic lookback selection identified a 30-day window as optimal, reflecting the relatively high volatility of sugar markets, where shorter windows allow the model to adapt more quickly to changing conditions. The volatility filter further improved the robustness of the strategy by avoiding the noisiest

periods — particularly the COVID shock in early 2020 and the sugar price rally in 2022–2023 — which significantly reduced drawdowns.

The final strategy produced a Sharpe ratio of 1.35, a maximum drawdown of -11.6%, and a Calmar ratio of 8.46 over the 2019–2024 period, with 10bps transaction costs included. These results indicate a strong risk-adjusted performance for a pairs trading strategy on commodities, combining relatively stable returns with controlled downside risk. The high win rate (83.8%) is consistent with a mean-reversion strategy, where profits are generated through frequent small gains rather than a few large trades.

If we were to go further, several improvements could be explored. The hedge ratio estimation could be enhanced using a Kalman filter instead of rolling OLS to increase adaptability. The entry and exit thresholds could also be optimized rather than fixed at ± 2 and 0. Finally, it would be interesting to test the strategy on other cointegrated pairs — such as KHC, PEP, and BG — to assess the robustness of the approach across different assets.

Final Conclusion

In this project, we worked on two different quantitative trading strategies applied to the sugar market. Even though the approaches are different, both are based on the idea that there are statistical relationships between the commodity and related assets that can be used to generate trading signals.

The pairs trading strategy developed in Game #2 shows that a relatively simple model can still deliver good results if it is well designed. By combining cointegration, a dynamic hedge ratio, and a volatility filter, we were able to build a strategy that performs well while keeping risk under control.

More generally, this project helped us understand how theoretical concepts like mean reversion or cointegration can actually be applied to real financial data. It also shows that having an economic intuition behind the model is important, not just relying on statistics.

Of course, the strategy could still be improved, for example by using more advanced models or testing it on other commodities. But overall, the results suggest that statistical arbitrage can be a relevant approach in commodity markets when it is carefully implemented.